Check for updates

International Journal of Sports Science & Coaching

Original research

A data-driven approach to assist offensive and defensive players in optimal decision making

International Journal of Sports Science & Coaching
2024, Vol. 19(1) 245–256
© The Author(s) 2023
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/17479541221149481
journals.sagepub.com/home/spo
Sage

Pegah Rahimian[1] and Laszlo Toka[2]

Abstract

Among all the popular sports, soccer is a relatively long-lasting game with a small number of goals per game. This renders the decision-making cumbersome, since it is not straightforward to evaluate the impact of in-game actions apart from goal scoring. Although several action valuation metrics and counterfactual reasoning have been proposed by researchers in recent years, assisting coaches in discovering the optimal actions in different situations of a soccer game has received little attention of soccer analytics. This work proposes the application of deep reinforcement learning on the event and tracking data of soccer matches to discover the most impactful actions at the interrupting point of a possession. Our optimization framework assists players and coaches in inspecting the optimal action, and on a higher level, we provide for the adjustment required for the teams in terms of their action frequencies in different pitch zones. The optimization results have different suggestions for offensive and defensive teams. For the offensive team, the optimal policy suggests more shots in half-spaces (i.e. long-distance shots). For the defending team, the optimal policy suggests that when locating in wings, defensive players should increase the frequency of fouls and ball outs rather than clearances, and when located in the centre, players should increase the frequency of clearances rather than fouls and ball outs.

Keywords

Ball possession, decision making, fouls, soccer analytics, team sport

Introduction

Recent breakthroughs in the applications of machine learning in soccer data have been contributed to soccer analytics to bridge the gap between coaching and science. The recently proposed action valuation methods, for example, Decroos et al.,[1] Liu and Schulte,[2] Liu et al.,[3] Fernandez et al.,[4] Alguacil et al.,[5] Fernandez and Born,[6] and Gyarmati and Stanojevic[7] help coaches and soccer practitioners to analyze the current performance of their team and players. However, the action valuation method leaves the coaches with the value of the performed actions, without any proper proposal of alternative actions, let alone the optimal one. To fill this gap, a few counterfactual reasoning methods, for example, Van Roy et al.,[8] and Sandholtz and Bornn[9,10] have been proposed that inspect the outcome of alternative decisions without the requirement of actual deployment in a match. Particularly, they provide answers to the questions such as *What would have happened if the team had modified the frequency of a particular action by x%?* However, their inspections are limited to a few alternative choices and do not explore the larger space of possibilities to get the optimal one.

In the meantime, the application of deep neural networks enables soccer data scientist to remedy the problem of high dimensional tracking data and offers the possibility to incorporate those in their analytics. In this work, we propose the application of deep reinforcement learning (RL) to leverage the high-dimensional tracking data, and to achieve the optimal offensive and defensive actions at the interrupting point of a particular possession. In our dataset, it is assumed that a possession starts from the

Reviewers: László Csató (Corvinus University of Budapest, Hungary)
Bill Gerrard (University of Leeds, UK)

[1]Department of Telecommunication and Media Informatics, Faculty of Electrical Engineering and Informatics, Budapest University of Technology and Economics, Budapest, Hungary
[2]MTA-BME Information Systems Research Group, Budapest, Hungary

Corresponding author:
Pegah Rahimian, Department of Telecommunication and Media Informatics, Faculty of Electrical Engineering and Informatics, Budapest University of Technology and Economics, Magyar tudósok körútja 2, Budapest 1111, Hungary.
Email: pegah.rahimian@edu.bme.hu

beginning of a deliberate on-ball action by a team, until it either ends due to some events such as ball out, foul, or shot from offensive players (regardless of who possesses the ball afterward, i.e., the next possession can belong to the same team or to the opponent), or interrupts by a defensive action of the opponent, such as pass interception, tackle, foul, out, or clearance.

We first analyze the actual offensive and defensive behavior of the teams in terms of the probability of performing any of the above-mentioned actions in those situations. We then use RL to tweak the actual probabilities and infer the optimal ones. Thus, given any possession of the offensive team, the action with the highest probability is the optimal action. Nevertheless, we face some challenges when designing the RL environment. First, soccer is a highly interactive and sparse rewarding game with a relatively large number of agents (i.e. players). Thus, state representation is challenging. Second, the spatiotemporal and sequential nature of soccer events and the dynamics of players' location on the field dramatically increase the state dimensions, which leads to the curse of dimensionality in machine learning tasks. Third, the game context in a soccer match severely affects the model prediction performance. Fourth, evaluating a trained optimal policy would require its deployment in a real soccer match. This work offers solutions to all these challenges. Sports professionals can use our policy optimization method after the match to check what actions the players performed, evaluate them, and find out the optimal alternative action in that situation. Therefore, we can evaluate the players and the teams in terms of their offensive and defensive strategies by inspecting how much their policies diverge from the optimal policy.

In summary, our work contains the following contributions:

- We propose an end-to-end RL framework that analyzes the event and tracking data and trains an optimal policy to discover the optimal offensive and defensive actions;
- We introduce a Markovian ball possession model, and a novel state representation to analyze the impact of actions on subsequent possessions;
- We propose a new reward function for each of the offensive and defensive actions, based on the output of the neural network predictor model;
- We provide interpretable results for soccer coaches in terms of the adjustments required in the frequency of offensive and defensive actions on different pitch zones.
- We rank teams' effectiveness in terms of their performance in selecting optimal offensive and defensive actions.

Related work

The state-of-the-art models in soccer analytics focus on several aspects such as evaluating actions, players, and strategies. A regression method on actions and shots was firstly proposed by McHale et al.[11] They estimate the number of shots as the function of crosses, dribbles, passes, clearances, etc. Coefficients show how important they are in generating shots. Another interesting player evaluation method is percentiles and player radars by StatsBomb.[12] This method estimates the relative rank for each player based on their actions. For example, a ranking can be assigned to a player for all their defensive actions (tackle/interception/block), their accurate passes, crosses, etc.

The application of a Markovian model in action valuation was first proposed by Rudd[13] and Keith.[14] The input of this model is the probability of ball location in the next five seconds. Assuming we have these probabilities, this model estimates the likely outcomes after many iterations based on the probabilities of transitioning from one state to another. Another application of the Markovian model is the Expected Threat (xT),[15] which uses simulations of soccer matches to assign values to actions. VAEP[1] is another action valuation model, which considers all types of actions; this model uses a classifier to estimate the probability that an action leads to a goal within the next 10 actions, and the game state is considered as three consecutive actions. This model ignores the concept of ball possession in valuation. In contrast, considering the possession, the Expected Possession Value (EPV) metrics in football by Fernandez and Born,[4] and in basketball by Cervone et al.[16] were proposed. These models assume a simple world in which the actions of the players inside possessions are limited to pass, shot, and dribble. Thus, these methods ignore any other actions such as foul, ball out, or errors, which frequently happen in critical situations.

Recently, researchers have utilized deep learning methods due to their promising performance in handling the spatiotemporal data. Fernandez and Born[6] present a convolutional neural network architecture that is capable of estimating full probability surfaces of potential passes in soccer. With regards to the application of RL in soccer analytics, a suitable soccer environment needs to be constructed, and players should be able to interact with this environment and get feedback (i.e. rewards) to gradually learn what is the best decision in each situation to achieve the best results (e.g. maximum number of goals) in the long run. Now such an environment can be a simulation of the games in which the players are allowed to interact with the environment until they learn the optimal actions. Such environments have been recently developed by a Google research team.[17] Furthermore, Mendes-Neves et al.[18] designed a data-driven simulation environment for assessing decision making in soccer. However, when it comes to analyzing a real soccer dataset and evaluating players based on their performance, we cannot use simulation and modify players' actions realistically. In such models, action valuations and policies must be learned from a fixed set of interactions and rewards that is sampled before the learning process. In this area, Liu and Schulte[2] took advantage of RL, by assigning value to

each of the actions in ice-hockey and soccer[3] using the Q-function. Moreover, Dick and Brefeld[19] used RL to rate player positioning in soccer. It was in the works by Rahimian et al.[20,21] that used RL to directly derive optimal policy rather than action valuation in soccer. In contrast to the latter two papers that aim to maximize the expected goals of the attacking teams, the current work assists both offensive and defensive players in terms of selecting the most impactful actions at the interrupting point of a possession.

Method

In this section, we provide technical details of modelling soccer dataset to derive the actual behavior of the teams in both offensive and defensive situations, designing the Markovian possession environment, and elaborating on the RL algorithm for the optimization process.

Dataset

The data used to conduct our experiments are collected by a company called InStat. The dataset includes both events and tracking information of 104 European soccer matches played in 11 different national leagues of Portugal, Russia, Hungary, and Germany in the 2017-2018 season. Each record of the event dataset describes one action performed by any of the attacker or defender players with additional features, such as action name (pass, shot, dribble, ball out, foul, clearance, tackle, interception, assist, and events such as goal, offside, own goal, challenges, etc.), (x,y) coordinates of the start and end location of the action, action result (successful or not), body id (action is performed by head, body, or foot), time second, player name, team name, opponents, and match id. The tracking data is one frame per second representation of positions for all players including home and away. We then merged tracking with event data. Each record of our merged dataset includes all players' coordinates with their corresponding features for each time-step of the event dataset. In total, the dataset includes 98,922 actions (including on-ball and defensive actions), 9462 possessions that are terminated by either one of the defensive actions of the opponent players or a shot, out, foul performed by the offensive players, and 329 goals.

Feature generation

We represent the game state by generating some features on top of tracking data. Thus the game state consists of 44-dimensional (x,y) normalized positions of the 22 players on the pitch from tracking data. Moreover, we construct angles and distances to goal, time remaining, home/away, and body id directly from event stream data. Table 1 shows the list of our constructed features on top of event and tracking data used for our machine learning tasks in the following sections.

Table 1. Feature set.

Feature name	Description
Angle to goal	The angle between the goal posts seen from the player location
Distance to goal	Euclidean distance from player location to center of the goal line
Time remaining	Time remained from action occurrence to the end of match half
Home/Away	Action is performed by home or away team?
Action result	Successful or unsuccessful
Body ID	Action is performed by head? body? foot?
Positions	44-dimensional exact locations (x,y) of opponents

Game state representation

One of the most challenging steps in soccer analytics is to represent the game state due to the high-dimensional structure of soccer dataset. We first specify the definition of ball possession. Due to the fluid nature of a game, it is not straightforward to have a comprehensive description of a possession, which applies to all different types of soccer logs provided by different companies (e.g. InStat, Wyscout, StatsBomb, Opta, etc.). In the InStat dataset and accordingly in this work, possessions for any home or away teams are clearly defined and numbered. A possession starts from the beginning of a deliberate on-ball action by a team, until it either ends due to some event like ball out, foul, or shot from offensive players (regardless of who possesses the ball afterward, that is, the next possession can belong to the same team or opposing team), or terminates by a defensive action of the opponent, such as pass interception, tackle, foul, out, or clearance. The possession can be transferred if and only if the team is not in the possession of the ball over two consecutive events. Thus, the unsuccessful touches of the opponent in fewer than three consecutive actions are not considered as a possession loss. Consequently, all actions of players of the same team (e.g. passes, shots, dribbles, crosses, assists, bad ball controls, fouls, challenges, free kicks, etc.) should be counted to get the possession length.

Now we can describe each of the offensive and defensive game states by generating the most relevant features and labels to them. We first filter the dataset for all possessions that are interrupted by a shot, foul, or ball out from offensive players, or a defensive action of the opponent players. The defensive actions are as follows: foul, clearance, out, tackle, and interception. We then demonstrate the state as the combination of different feature vectors X (cf. Tables 1), and one-hot representation of the action A for all the actions inside each of the constructed attacking possessions, excluding the interrupting action. Thus, the varying possession length is the number of actions within a possession, excluding the ending one. Then, the state is a 2-dimensional array, with the first dimension being the possession length (varying for each possession), and the

second dimension being the number of features. Therefore, a mth state/possession with length n can be represented as $S_m = [[X_0, A_0], [X_1, A_1], \ldots, [X_{n-1}, A_{n-1}]]$ and we aim to predict the interrupting defensive action A_n.

Inferring offensive and defensive players' actual behavior

We aim to discover the actual offensive and defensive behavior of the soccer teams in each situation. Particularly, we seek for the probability of each of the offensive and defensive action types that a player can perform at the end of the possession. To do so, we employ a neural network to estimate the probabilities and predict the actions according to the historical games in the dataset. In the soccer event dataset, each possession is represented by a sequence of actions. We aim to classify these possessions based on the offensive and defensive action that interrupted them. Thus, each possession is terminated by one of the following classes: shot, foul, ball out performed by an offensive player, or tackle, interception, clearance, foul, or ball out performed by a defensive player. We utilize the classification capability of sequence prediction methods by training a CNN-LSTM network,[22] using CNN for spatial feature extraction of input possessions, and LSTM layer with 100 memory units to support sequence prediction and to interpret features across time steps. Figure 1 depicts the architecture of our network. Note that the feature vector X has a fixed length for each individual action, but varying for all actions in the state (because possession length or the number of actions varies). This is one of the main challenges in our work as most machine-learning methods require fixed-length feature vectors. In order to address the challenge of dynamic length of possession features (second dimension of the possession input array), we use truncating and padding. We limit the total number of actions in a possession to 10, truncating long possessions and we pad the short possessions with zero values. In this case, we will have a fixed length of sequences through the whole dataset for modeling. The limitation of the maximum number of actions in a possession is one of the parameters of our approach that we empirically set it to 10 after network calibration and sensitivity analysis. In our dataset, only 2.3% of the attacking possessions were truncated due to their length exceeding 10 actions.

Consequently, the network estimates the categorical probability distribution over the interrupting offensive and defensive actions for any given possession.

Markov decision process

So far, we could model the actual behavior of the soccer teams in terms of the probability of interrupting the possession of the attacking team by performing any of the aforementioned offensive and defensive actions, given any game state described in the previous section. However, the estimated probabilities are not sufficient to assist players and coaches in optimal decision making. This is due to the fact that the neural network is trained with the data of historical games (i.e. the actual actions that the player performed to interrupt the opponent possession). However, it is possible that other actions would have been better for attacking or defending. Thus, RL can help to learn from the short-term rewards that the attackers and defenders earned in the historical games, to increase the probability of highly rewarding actions, and to decrease the low rewarding ones when the player gets in the same situation in the future.

In order to design and train a RL model, we first represent a soccer game as a Markov Decision Process (MDP). An MDP consists of the following components:

- State: which describes the situation in the environment;
- Action: which specifies one of the set of possible actions;
- Policy: which determines the probability of selecting each action in the states;
- Reward: which determines the short-term feedback from environment after performing the action.

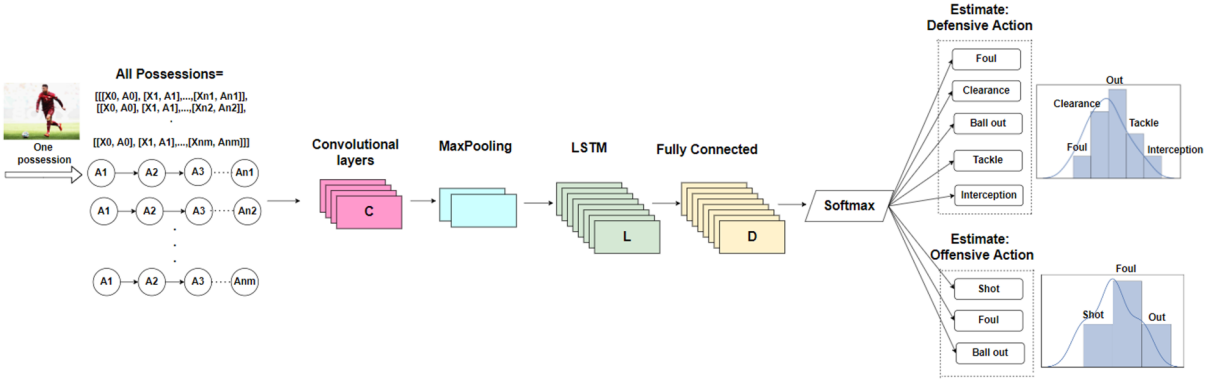


Figure 1. Convolutional neural network and long short-term memory (CNN-LSTM) network structure for prediction of the interrupting offensive and defensive actions, given a possession. Input possessions represent both state features vector (X), and one-hot vector of actions (a), excluding the ending action. There are m possessions in the dataset, with varying lengths of $\{n_1, n_2, \ldots, n_m.\}$. The output is the categorical distribution of each offensive and defensive action at the end of the possession.

Next, we define each of these components in our soccer setup.

Our designed MDP consists of the following components:

- State (s): Sequence of actions and their features in one interrupted possession of the attacking team, excluding the ending action.
- Action (a): The offensive or defensive action of the players that resulted in possession interruption. In this work, we restrict the defensive actions to foul, clearance, out, tackle, and interception, and offensive ones to shot, foul, ball out.
- Episode (τ): Sequence of possessions of the attacking team, until they score a goal (concede a goal for defending team), or match half ends.
- Reward ($r(s_t, a_t)$): Reward acquired from each ending action by the attacking or defending player at the end of a possession.
- Objective Function: Sum of discounted rewards through all episodes of the games. Intuitively, the attacking team aims to maximize the expected goal and prevent possession loss. The defending team aims to minimize the expected goal of the attacking team and interrupt their possession, which favors the defending team.

Reward function

Owing to the complex and sparse reward environment of soccer games, it is not straightforward to assign rewards to the actions. Several works (e.g. Liu and Schulte[2] and Liu et al.[3]) propose 0-1 rewarding systems, which assign 1 to the goal actions, and 0 to the rest of the actions. However, this sparse rewarding system is not applicable in our case, since we want to derive the optimal *offensive and defensive* actions with different objective functions. Therefore, we propose hand-crafted reward functions which assign rewards to each of the offensive and defensive actions separately according to their impact on maximizing or minimizing scoring chances at the end of the episode. For both reward functions, we need to estimate the probability of scoring a goal for the ball possessor team within the next 10 actions of the interrupted point of possession, before (s_{i-1}) and after (s_i) action a is committed.

Offensive team. For any of the offensive ending actions (i.e. shot, foul, or out), the ball possession could either be saved, or be transferred to the opponent team. Depending on the type of ending action and if it could save the possession or not, we assign reward to each of the action as follows:

$$r(s, a) = \begin{cases} xG & \text{if a is a shot;} \\ P_{score(s_i)} - P_{score(s_{i-1})} & \text{else and possession saved} \\ -P_{score(s_{i-1})} & \text{otherwise,} \end{cases} \quad (1)$$

where xG is the expected goal of the shot (i.e. probability of scoring a goal from a particular shot given the shot features), and $P_{score(s_i)}$ and $P_{score(s_{i-1})}$ are probabilities of scoring a goal within the next 10 actions of the interrupted point of the possession, after (s_i) and before (s_{i-1}) the action a is committed. Intuitively, if the action is a shot, we assign the value of its expected goal as the reward. If the action is not a shot, but it could save the possession for the attacking team (e.g. they could retake the possession within three consecutive actions), we calculate the difference between the probability of scoring within the next 10 actions before and after action a is committed. Thus, the action that increases the probability of scoring should be highly rewarded and vice versa. Finally, an action that resulted in possession loss is punished by the negative value of the goal scoring probability before the ending action, since the offensive team lost this opportunity.

Defensive team. Analyzing defensive actions are more challenging than offensive actions since they usually prevent something from happening (e.g. the threat of goal scoring by the offensive team). The probability of scoring before the ending action ($P_{score(s_{i-1})}$) is considered as a threat for the defensive team and they need to prevent it by either interrupting the possession or performing an ending action (e.g. out or foul) which decreases that threat in the subsequent possession. Similar to offensive ending actions, the defensive ending action could be either successful (i.e. interrupt the attacking possession), or unsuccessful (possession retains for the attacking team). Therefore, for a successful defensive action, we assign a positive reward equal to the value of the probability of the attacking team's goal scoring $P_{score(s_{i-1})}$ since the successful defensive action prevented this threat, and for unsuccessful ones, we estimate the change in the probability of the attacking team scoring before and after the defensive action is committed. Thus, if the defensive action resulted in decreasing the threat, we assign a positive reward, and if it resulted in increasing the threat, we punish it with negative rewards. The mathematical formulation of the reward $r(s, a)$ assigned to the defensive action a given state s is provided in (2) as follows:

$$r(s, a) = \begin{cases} P_{score(s_{i-1})} & \text{if a is successful;} \\ P_{score(s_{i-1})} - P_{score(s_i)} & \text{otherwise.} \end{cases} \quad (2)$$

Therefore, a large positive value of $P_{score(s_{i-1})} - P_{score(s_i)}$ indicates the better effect of the defensive action to prevent goal scoring withing the next 10 actions, and such a defensive action should be highly rewarded, and vice-versa.

Training protocol and return

For each game state, the network needs to decide about performing the appropriate offensive and defensive action with the corresponding parameter gradient. The parameter gradient tells us how the network should modify the parameters if we want to encourage that decision in that possession in the future. We modulate the loss for each interrupting action

taken at the end of a possession according to their eventual outcome since we aim to increase the log probability of successful actions (with higher rewards) and decrease it for the unsuccessful actions.

We define discounted reward (return) for episode τ in equation (3):

$$R(\tau) = \frac{\sum_{t=0}^{\infty} \gamma^t r(s_{\tau,t}, a_{\tau,t})}{\sum_{t=0}^{\infty} \gamma^t}, \quad (3)$$

where γ is a discount factor (set to 0.99 in this work), and r is the estimated reward for time-step t (t^{th} possession in the episode) after standardization to control the gradient estimator variance. R shows that the strength of encouraging a sample action at the end of a possession is the weighted sum of rewards (expected goals). In this work, we constrain the look ahead to the end of the episodes, which is either goal scoring or end of a match half.

Policy gradient

So far, we are able to predict the most probable offensive and defensive actions that an average team will take to either save the possession for the offensive team in case of attack, or interrupt the offensive possession and prevent the threat in case of defense. We propose employing CNN-LSTM network that is able to predict the probabilities of performing each of the defensive actions. Moreover, by predicting the probability of goal scoring withing the next 10 actions before and after the occurrence of the ending action, we can estimate the rewards described in the previous section. Therefore, each of the ending actions is assigned positive or negative rewards depending on their impact. It might happen that a defensive action, such as committing a foul, has temporarily interrupted the possession and led to a free kick with a large expected goal, or a clearance that moved the ball to an area on the pitch that opened up space for the attacking team. In order to decrease the likelihood of such actions with high negative rewards in the same situation in the future, and to increase the likelihood of actions that can earn a large positive reward in the same situation, we propose training the network with Policy Gradient (PG) which is a type of score function gradient estimator. Using PG, we aim to train a defensive policy network that directly learns the optimal offensive and defensive policies by learning a function that outputs the best offensive and defensive actions to be taken at the end of the possession with a maximum impact on the future depending on the objective function of the team.

We seek to learn how the distribution of the offensive and defensive actions should be shifted (through its parameter θ), in order to increase the reward of the taken actions.

Using PG, the gradient vector computes a direction in the parameter space leading to an increase in the probability assigned to the action a. Consequently, high rewarding actions will tug on the probability density stronger than low rewarding actions. Therefore, by training the network, the probability density would shift around in the direction of high rewarding actions, making them more likely to occur in future. See Figure 2 for the workflow of the PG algorithm for the defensive actions.

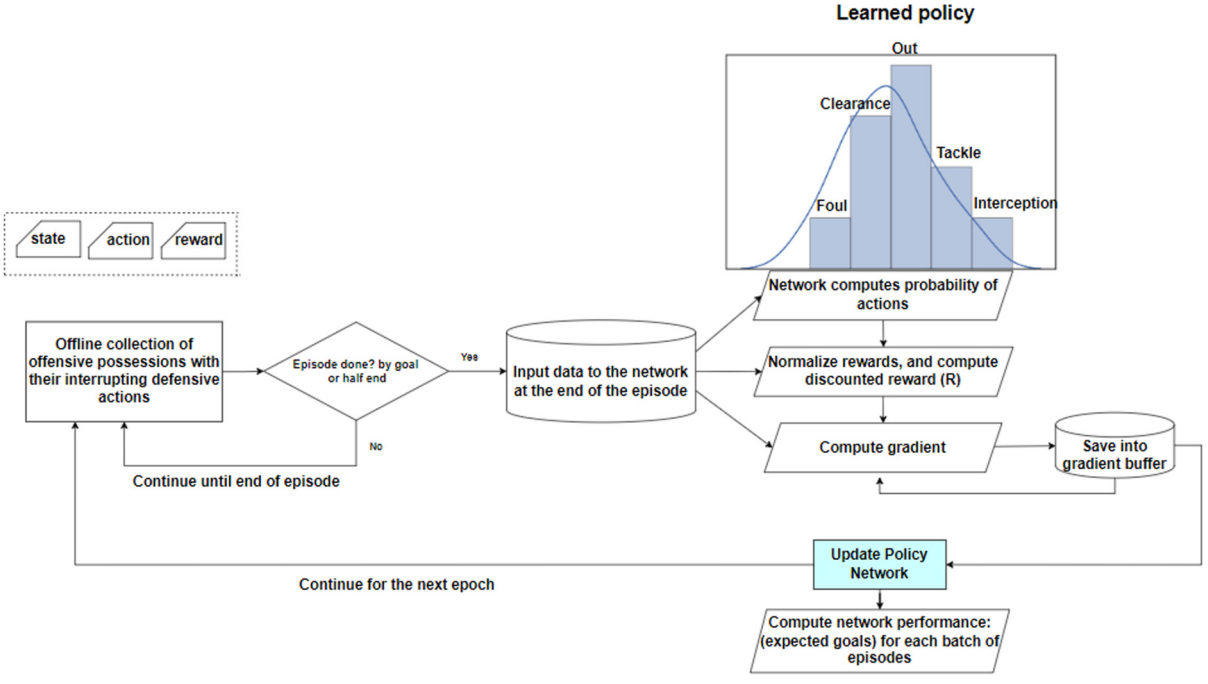


Figure 2. Policy gradient workflow.

Results

In this section, we provide the experimental results and evaluation of the proposed approach in different phases of analysis.

Offensive and defensive behavior prediction results

In this section, we inspect the performance of the teams' offensive and defensive behavior prediction (i.e. the probability of interrupting a possession by each of the offensive and defensive actions. In order to handle the spatiotemporal nature of our dataset, we needed a sophisticated model and rich feature set, which could optimize the prediction performance. Thus, model selection was the core task of this study. We first created appropriate state dimensions suitable for each model by reshaping the state inputs, then fed our reshaped arrays to the following networks: 3D-CNN, LSTM, Autoencoder-LSTM, and CNN-LSTM, to compare their classification performance (cf. Table 2). We chronologically split the attacking possessions into 80% of train, 10% of validation for hyperparameter tuning and model selection, and the remaining 10% as hold-out data for testing. Categorical cross-entropy loss on train and validation datasets is used to evaluate the model. All layers of the networks are carefully calibrated. Dropout is used to reduce interdependent learning among units, and early stopping is used to avoid overfitting. As the table suggests, CNN-LSTM[22] outperforms other models in terms of accuracy and loss in the test set.

Expected goal results

In order to estimate the reward functions in (1) and (2), we need to calculate the probability of scoring a goal by the attacking team within the next 10 actions of the point of the interruption, before and after the interrupting action. To do so, we again collect all possessions described before, and label them as a goal or no-goal depending on what has happened within the next 10 actions considering all features listed in Table 1, including the locations. This binary classification model was trained with the same CNN-LSTM network and showed an accuracy of 78% on the test set.

Optimization results

In contrast to the behavior prediction and the expected goal models in which we have access to the labels, it is a challenging task to evaluate the optimization results, since there is no ground truth method for action valuation or optimal actions in soccer. Therefore, we evaluate the performance of our proposed framework with an eye towards two questions: (1) How well our optimized network maximizes the expected goal for an offensive team and minimizes the threat for a defensive team in comparison to the actual policy? We answer this question by the off-policy evaluation (OPE) method. (2) What is the intuition behind the selected actions of our target policy? We elaborate on this by providing two scenarios of the different situations in a particular match from the dataset.

Off-policy evaluation with importance sampling. Applying the off-policy method in our soccer analysis problem, we faced the following challenge: while training can be performed without a real robot (simulator), the evaluation cannot, because we cannot deploy the learned policy in a real soccer match to test its performance. This challenge motivated us to use OPE, which is a common technique for testing the performance of a new policy, when the environment is not available or it is expensive to use. With OPE, we aim to estimate the value and performance of our newly optimized policy based on the historical match data collected by a different actual policy obeyed by the players. For this aim, we use, the importance sampling method, similar to related works, such as Xie et al.[23] The importance sampling method takes samples from the actual policy distribution to evaluate the performance of the target policy distribution. The workflow of the evaluation with importance sampling for defensive actions is sketched in Figure 3.

Discovering the optimal actions in different pitch zones

Our proposed framework tweaks the probability distribution of occurring offensive and defensive actions given any possession. Thus, the action that has the maximum

Table 2. Classification performance of different design choices for spatiotemporal analysis. (Inference times are the average running times over 20 iterations of training using a server enriched with a Tesla K80 GPU). The dimension of the state is calculated as follows: [features (6) + locations (44) + actions (11)].

State dimension	Spatiotemporal model	Accuracy (%)	Loss	Inference time (s)	Parameters
[9462,10,61]	3D-CNN	70	0.65	2.18	122,411
	LSTM	66	0.71	0.98	192,523
	Autoencoder-LSTM	76	0.65	2.29	321,192
	CNN-LSTM	**80**	**0.55**	1.39	201,019

3D-CNN: three-dimensional convolutional neural network; LSTM: long short-term memory; CNN-LSTM: convolutional neural network and long short-term memory. The best performing model is bold in the table.

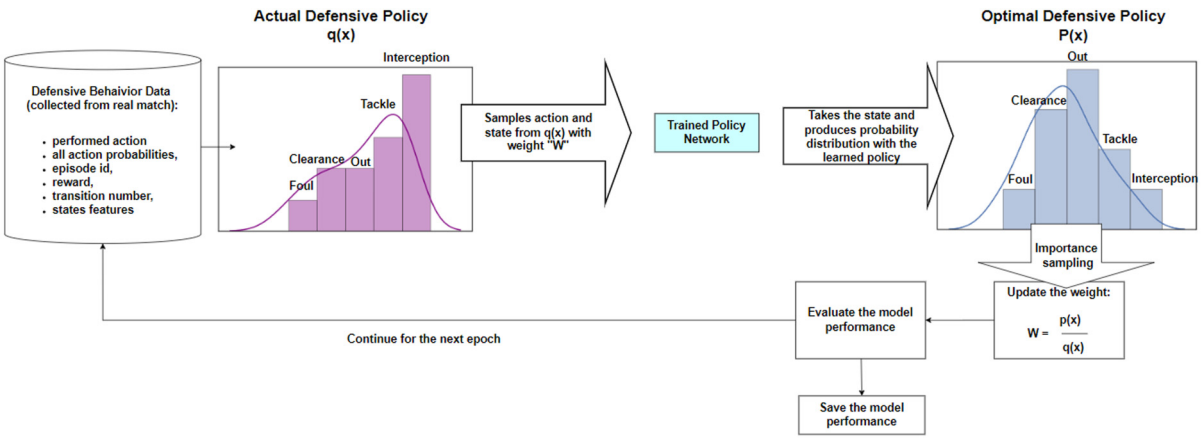


Figure 3. Importance sampling workflow to evaluate the performance of the optimal policy.

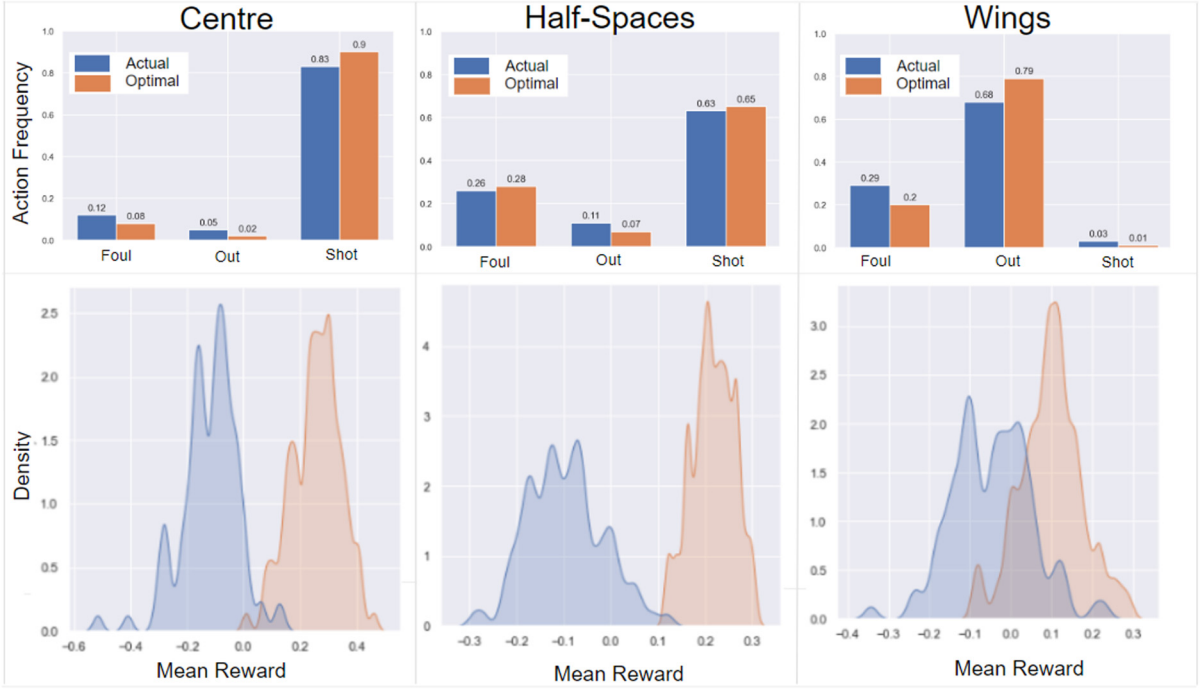


Figure 4. Optimization results for offensive actions in different pitch zones. The top charts illustrate the actual and optimal offensive action frequencies and the bottom charts show the Kernel Density Estimation (KDE) of the mean rewards following each of the actual and optimal policies. Mean rewards show the amount of expected goal difference the offensive team can gain on an average of all games following each of the actual and optimal policies.

probability after optimization is considered as the optimal action. This application assists soccer data analysts to feed any kind of states of the possessions to the policy networks and infer the optimal offensive and defensive actions. However, coaches are more enthusiastic with high-level results that help them in optimal decision making and planning for future games. To create such insights, we divide the soccer pitch into five different zones: left wing, left half-space (i.e. the area between left wing and center), center, right half-space (i.e. the area between right wing and center), and right wing. We then compare the action frequency using the actual policy (computed from the event data) and our trained optimal policy in each of the zones for each of the offensive and defensive teams. Figures 4 and 5 illustrate the comparison of action frequency in each of the zones following actual and optimal policies. The top row in these figures provides the exact action frequencies, and the bottom rows show the Kernel Density Estimation (KDE) of the mean rewards on average of all the 104 games following actual and optimal policies. The

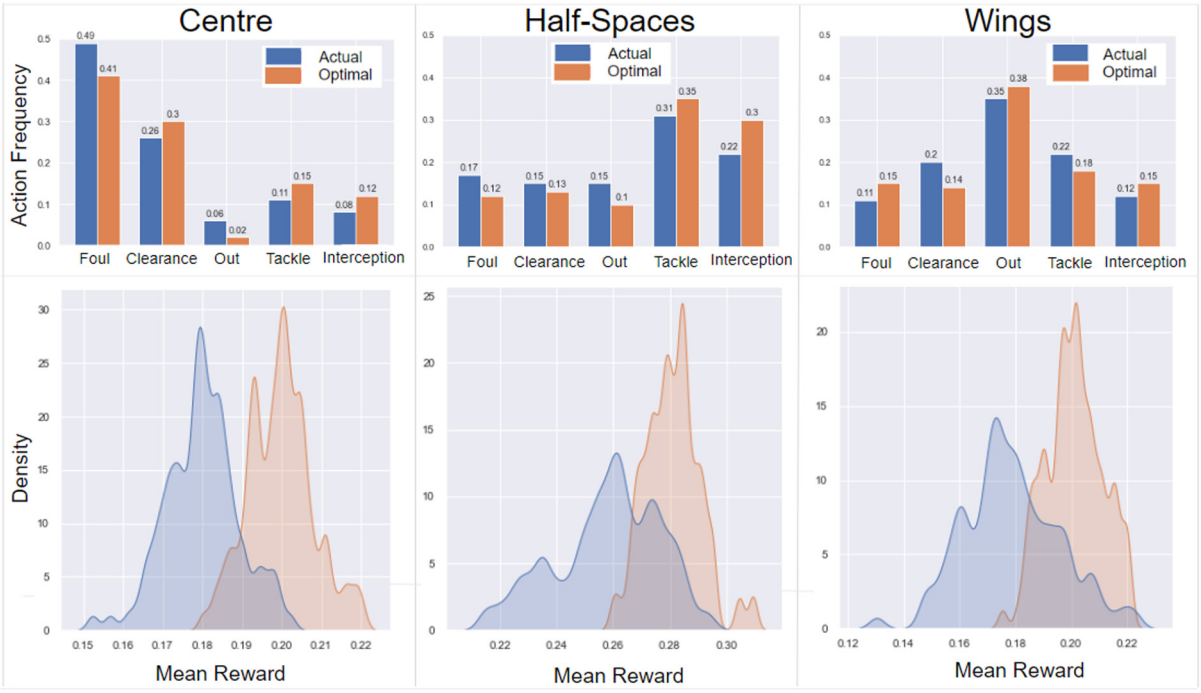


Figure 5. Optimization results for defensive actions in different pitch zones. The top charts illustrate the actual and optimal offensive action frequencies and the bottom charts show the Kernel Density Estimation (KDE) of the mean rewards following each of the actual and optimal policies. Mean rewards show the amount of decrease in the chance of scoring by attacking teams (preventing the threat) before and after the defensive action of the defender team on an average of all games. Thus, a higher mean rewards favours the defending team.

summary of the most important results for the offensive and defensive approaches according to their objectives and reward functions is as follows (Note: we provide some justification of the obvious results in soccer games in parentheses that partially proves the validity of our approach):

Offensive approach

- In Centre: the optimal policy proposes more shots (due to the large xG values in this zone) and less fouls (due to the punishment resulting in a penalty or free kick). The proposed modification in action frequencies results in a 0.4 increase in mean rewards (i.e. expected goal of the attacking team) from −0.1 to 0.3 on average of all teams.
- In Half-Spaces: the optimal policy suggests more fouls and more shots (i.e. long-distance shots). The proposed modifications in action frequencies results in a 0.3 increase in mean rewards from −0.1 to 0.2 on average of all teams.
- In Wings: the optimal policy suggests increasing the number of ball-out actions and committing less fouls. This modification in frequencies changes the mean rewards from −0.1 to 0.15 on average of all the games.

In total, the amount of expected goal difference before and after the interrupting action for the offensive team would be −0.1 (i.e. they would decrease the expected goal by 0.1 by performing the interrupting action) following the actual policy, and they would be able to increase their mean rewards to 0.22 (i.e. they would increase the expected goal by 0.22 by performing the interrupting action) following the optimal policy on average of all episodes of the games, which indicates a 0.32 improvement in the expected goal difference. This improvement is relatively large that is due to the large value of the xG (the expected goal of a shot action) in the reward function of the offensive team, and the higher frequency of the shot actions in the optimal policy.

Now, we consider an offensive scenario to see how the optimized policy works compared to the actual policy.

- **Scenario 1:** Offensive team's player missed goal scoring opportunity: Figure 6 shows the 24th episode of the match. Player A from the offensive team stops a long sequence of passes by committing a foul and he gets a reward of −0.16. In this second, there is high pressure from the defending team players (B, C, and D). So A tries a foul to prevent possession loss. Then player D from the defending team gets the ball. The policy network suggests shooting the ball instead of committing the foul. So player A could gain a reward (expected goal) of 0.16, meaning that the probability of goal scoring is 0.16, and he missed this opportunity.

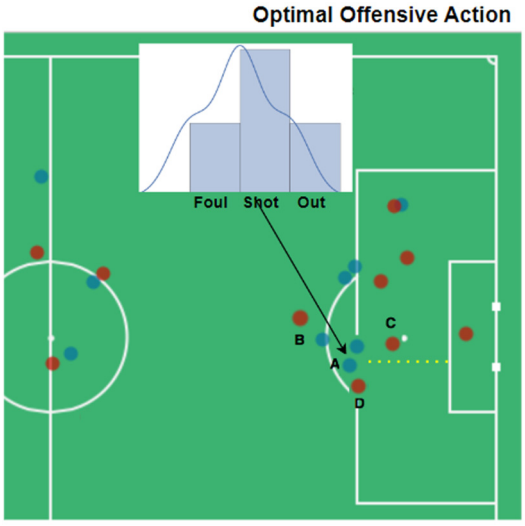


Figure 6. Offensive scenario: Blue dots are offensive team players, and reds are defenders. Player (a) is the attacker and the ball holder, and players (b, c, and d) are the defenders. The attack direction is from left to right. Yellow dashed lines show the optimal trajectory of ball, if player (a) was following optimal policy. The probability distributions show the optimal output of our trained policy network.

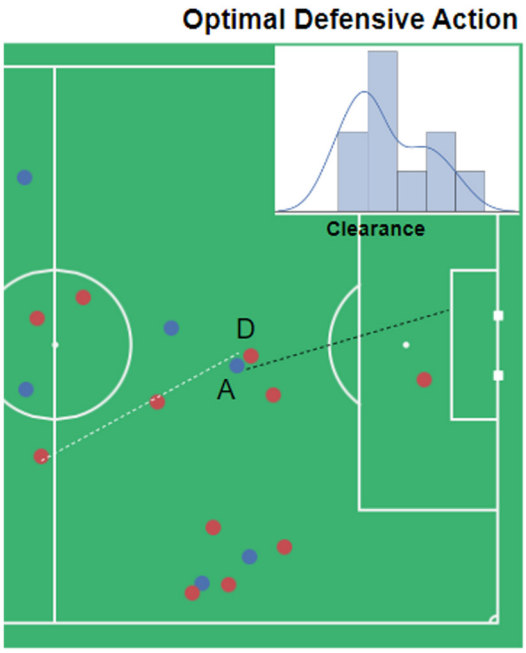


Figure 7. Defensive scenario: Blue dots are offensive team players, and reds are defenders. Player (a) is the attacker and the ball holder, and players (d) are is the defender. The attack direction in is from left to right. Black dashed line shows the actual ball trajectory and White dashed lines show the optimal trajectory of ball, if player (d) was following optimal policy. The probability distributions show the optimal output of our trained policy network.

Defensive approach

- In Centre: the optimal policy proposes less fouls (since it usually results in a free kick and high chances of goal by attackers), more clearances, less ball outs, more tackles, and interceptions. The proposed modifications in defensive action frequencies result in a 0.02 increase in mean rewards for the defending team (i.e. 0.02 decrease in the scoring chance of the attacking team) from 0.18 to 0.20 on average of all games.
- In Half-Spaces: the optimal policy suggests less fouls, clearances, and ball outs, and more tackles and interceptions instead. The proposed modifications in action frequencies result again in a 0.02 increase in mean rewards for the defending team (i.e. 0.02 decrease in the scoring chance of the attacking team) from 0.26 to 0.28 on an average of all games.
- In Wings: the optimal policy suggests increasing the number of fouls (since it usually results in a free kick which is less riskier from the wings than in the center), less clearance, more ball outs (since it usually crosses the touchline and results in a throw-in which is considered to be less riskier than a free kick), less tackles, and more interceptions. The proposed modifications in action frequencies result in a 0.03 increase in mean rewards for the defending team (i.e. 0.03 of decrease in the scoring chance of attacking team) from 0.17 to 0.20 on average of all games.

In total, the defensive team can decrease the expected goal difference of the offensive team by 0.2 following the interrupting defensive action of the actual policy, and they can increase this amount to 0.22 following optimal policy, which indicates a 0.02 improvement in decreasing the scoring probabilities on average of all episodes through the games.

Now, we consider a defensive scenario to see how the optimized policy works compared to the actual policy.

Scenario 2: Figure 7 shows the 52nd episode of the match. Player A from the attacking team holds the ball in the center where there is a high chance of the expected goal (0.2). Player D from the defending team tries to stop his attack and commits a foul, which results in a risky free-kick quickly performed by the attacking team (due to the low number of defenders in the box), and they score a goal. Our method assigns −0.03 reward to the action of player D, since this action resulted in an increase of 0.03 in the probability of goal scoring by the attackers within the next 10 actions from 0.2 to 0.23. However, our trained policy network suggests a clearance instead of the foul for player D with which the goal scoring could have been prevented, and they could have sent the ball to the other side of the pitch.

Measuring teams' effectiveness

Measuring teams' effectiveness is one of the most prominent use-cases of our approach. Since we access both the actual

and optimal probabilities of performing any offensive and defensive actions for each team in different situations of the soccer games, we could estimate their distances and use it to measure effectiveness. To do so, we use Chi-square distance that is a statistical method, generally used for measuring similarity between histograms. The Chi-square distance of two arrays "x" and "y" with "n" dimension is calculated as follows:

$$\chi^2 = \sum_{i=1}^{n} \frac{(x_i - y_i)^2}{(x_i + y_i)} \tag{4}$$

In our case, x and y arrays are the probabilities of the actions following actual and optimal policies, respectively. n is the number of our discrete action types that is three in case of offense (i.e. foul, out, shot), and five in case of defense (i.e. foul, clearance, out, tackle, interception). For each team, we calculate the Chi-square distance for each of the offensive and defensive actions that the players performed, normalize it, sum them up and divide it by the total number of performed offensive and defensive actions separately. As the result, a smaller distance between probabilities shows better effectiveness for a team. Therefore, the similarity or effectiveness of the teams is calculated as follows:

$$\text{Effectiveness} = 1 - \left(\frac{1}{\mathrm{m}} \sum_{\mathrm{j}=1}^{\mathrm{m}} \frac{\chi_{\mathrm{j}}^2}{\mathrm{n}} \right); \tag{5}$$

where χ_j^2 is the χ^2 of the jth offensive or defensive action that players performed, and m is the total number of offensive and defensive actions performed by the teams. As the result, the larger effectiveness of a team shows their better performance in selecting optimal actions in any of the offensive and defensive situations. Table 3 ranks the 11 teams in our dataset according to their effectiveness in offensive and defensive situations separately. According to the table, team A has the largest effectiveness in offensive actions (i.e. they mostly tend to select the optimal offensive actions), whereas the largest effectiveness in defensive actions belongs to team E (i.e. they mostly tend to select the optimal defensive actions). Similarly, team K and team I showed the worst effectiveness in offensive and defensive actions, respectively.

Conclusion

This work proposes a data-driven deep reinforcement learning framework to help offensive and defensive players to select the best action at the interrupting point of a possession.

Our framework built on a training policy network helps the players and coaches to compare their actual policy with the optimal one. More specifically, sports professionals can feed any state with the proposed possession features and state representation to find the optimal offensive and defensive actions. Furthermore, coaches can use the results of our optimization for future planing in terms of the adjustment required in action frequencies on different pitch zones. We analyzed 104 matches from 11 teams and showed that the optimal policy network can increase their mean rewards on average of all games in the dataset at both offensive and defensive situations. Concisely, the optimization results have different suggestions for offensive and defensive teams. For offensive team, the optimal policy suggests more shots in half-spaces (i.e. long-distance shots). For the defending team, the optimal policy suggests that when locating in wings, defensive players should increase the frequency of fouls and ball outs rather than clearances, and = when located in the center, players should increase the frequency of clearances rather than fouls and ball outs. Moreover, we provide the studied teams' ranking in terms of their effectiveness in selecting optimal offensive and defensive actions throughout their games.

In this work, we optimize the policies independently and assume that the behavior of the opponent team is not adapted to the optimal policy. Thus, a direction for future

Table 3. Teams ranking according to their effectiveness and selecting optimal actions. Teams are sorted according to their effectiveness in offensive actions.

	Offense: $n = 3$			Defense: $n = 5$		
Teams	$\sum \frac{\chi^2}{n}$	Number of actions (m)	Effectiveness	$\sum \frac{\chi^2}{n}$	Number of actions (m)	Effectiveness
A	101.21	349	0.71	161.51	521	0.69
B	99.51	321	0.69	162.54	602	0.73
C	153.45	495	0.69	119.40	398	0.70
D	112.71	289	0.61	166.46	406	0.59
E	139.81	341	0.59	115.06	523	0.78
F	162.96	388	0.58	156.48	489	0.68
G	134.82	321	0.58	388.22	658	0.41
H	255.51	501	0.49	119.14	322	0.63
I	221.88	422	0.46	244.61	401	0.39
J	194.88	348	0.44	312.12	612	0.49
K	148.09	251	0.41	241.92	504	0.52

work is to use multi-agent reinforcement learning to adapt the opponent teams with the optimal policy as well, which might decrease the amount of improvement in mean rewards reported in this article. Another limitation of the current study is the focus on on-ball defensive actions, hence, the ignorance of the positioning decisions of the teams. Moreover, further information of the game context, such as different attributes of passes (e.g. height, length, angle, etc.), teams' and players' abilities and their specific strategies would significantly improve the results.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship and/or publication of this article: Project no. 128233 has been implemented with the support provided by the Ministry of Innovation and Technology of Hungary from the National Research, Development and Innovation Fund, financed under the FK_18 funding scheme.

ORCID iDs

Pegah Rahimian https://orcid.org/0000-0002-0293-2739
Laszlo Toka https://orcid.org/0000-0003-1045-9205

References

1. Decroos T, Bransen L, Van Haaren J et al. Actions speak louder than goals: Valuing player actions in soccer. In: *ACM KDD 2019*. ACM.
2. Liu G and Schulte O. Deep reinforcement learning in ice hockey for context-aware player evaluation. In: *IJCAI 2018*.
3. Liu G, Luo Y, Schulte O et al. Deep soccer analytics: Learning an action-value function for evaluating soccer players. *Data Min Knowl Discovery* 2020; 34: 1531–1559.
4. Fernandez J, Bornn L and Cervone D. Decomposing the immeasurable sport: A deep learning expected possession value framework for soccer. In: *MIT Sloan Sports Analytics Conference 2019*.
5. Alguacil FP, Fernandez J, Arce PP et al. Seeing in to the future: Using self-propelled particle models to aid player decision-making in soccer. In: *MIT Sloan Sports Analytics Conference 2020*.
6. Fernandez J and Born L. Soccermap: A deep learning architecture for visually-interpretable analysis in soccer. In: *ECML PKDD 2020*.
7. Gyarmati L and Stanojevic R. Qpass: A merit-based evaluation of soccer passes. In: *KDD Workshop on Large-Scale Sports Analytics 2016*.
8. Van Roy M, Robberechts P, Yang WC et al. Leaving goals on the pitch: Evaluating decision making in soccer. In: *MIT Sloan Sports Analytics Conference 2021*.
9. Sandholtz N and Bornn L. Replaying the NBA. In: *MIT Sloan Sports Analytics Conference 2018*.
10. Sandholtz N and Bornn L. Markov decision processes with dynamic transition probabilities: An analysis of shooting strategies in basketball. *Ann Appl Stat* 2020; 14: 1122–1145.
11. McHale IG, Scarf PA and Folker DE. On the development of a soccer player performance rating system for the English Premier League. *Interfaces* 2012; 42: 339–351.
12. StatsBomb. New data, new statsbomb radars. https://statsbomb.com/2018/08/new-data-new-statsbomb-radars/, 2018.
13. Rudd S. A framework for tactical analysis and individual offensive production assessment in soccer using Markov chains. http://nessis.org/nessis11/rudd.pdf, 2018.
14. Keith G. A Markov model of football: Using stochastic processes to model a football drive. *J Quant Anal Sports* 2012; 8: 9–9.
15. Sing K. Introducing expected threat (xt) modelling team behaviour in possession to gain a deeper understanding of buildup play. https://karun.in/blog/expected-threat.html, 2018.
16. Cervone D, Amour A, Bornn L et al. Pointwise: Predicting points and valuing decisions in real time with NBA optical tracking data. In: *MIT Sloan Sports Analytics Conference 2014*.
17. Kurach K, Raichuk A, Stanczyk P et al. Google research football: A novel reinforcement learning environment. *CoRR* 2019; abs/1907.11180. doi:10.48550/arXiv.1907.11180.
18. Mendes-Neves T, Mendes-Moreira J and Rossetti RJF. A data-driven simulator for assessing decision-making in soccer. In: *EPIA Conference on Artificial Intelligence 2021*.
19. Dick U and Brefeld U. Learning to rate player positioning in soccer. *Big Data* 2019; 7: 71–82.
20. Rahimian P, Oroojlooy A and Toka L. Towards optimized actions in critical situations of soccer games with deep reinforcement learning. In: *DSAA 2021*.
21. Rahimian P, VanHaaren J, Abzhanova T et al. Beyond action valuation: A deep reinforcement learning framework for optimizing player decisions in soccer. In: *MIT Sloan Sports Analytics Conference 2022*.
22. Donahue J, Hendricks LA, Rohrbach M et al. Long-term recurrent convolutional networks for visual recognition and description. In: *IEEE Conference on Computer Vision and Pattern Recognition 2016*.
23. Xie T, Ma Y and Wang Y. Optimal off-policy evaluation for reinforcement learning with marginalized importance sampling. *CoRR* 2019.